

NIMBLIZE

Future Architecture Recommendations, Scaling Roadmap, and Long-Term AI Evolution for Nimblize

Nimblize Future Implementation & Production Roadmap

This document contains future implementation recommendations, scaling strategies, and proposed enhancements beyond the currently implemented Phase 4 scope.

Organization: Nimblize

Domain: AI & Automation

Domain Leader: Aastha Shukla

CTO & Co-Founder: Anshul Sinha

Version: 4.2.0-PROD

Date: July 2026

Classification: Production-Ready Engineering Blueprint

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Nimblize Phase 4: Production Implementation of AI & Automation Architecture

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Future Architecture Recommendations, Scaling Roadmap & AI Evolution
Domain: AI & Automation

Co

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Executive Summary

Business Problem: Nimblize operates across two market vectors — a B2B Growth Ecosystem that delivers automated SEO intelligence, competitor diagnostics, and affiliate commerce analytics, and a B2C Product Recommendation Engine requiring sub-15ms semantic search. Both divisions previously relied on manual analysis workflows that could not scale. The competitive intelligence cycle — from raw web scraping to actionable dashboard insights — averaged 4–6 hours of human labor per competitor profile, creating an operational bottleneck that directly limited the company's market coverage.

Technical Problem: Existing approaches to automating this workflow using single-prompt LLM inference proved structurally unreliable. When processing long, unstructured competitor web pages, monolithic prompts exhibited severe context drift, with hallucination rates exceeding 18% in early benchmarks. These hallucinations manifested as fabricated traffic metrics, invented affiliate networks, and keyword lists that bore no relationship to the source text. Furthermore, there was no programmatic mechanism to validate LLM outputs before they entered the production database, creating a data integrity risk that disqualified single-prompt approaches from production use.

Solution Developed: A decoupled Multi-Agent Topology orchestrated by a LangGraph state machine, combined with a Hierarchical Parent-Child RAG pipeline and an algorithmic Confidence Evaluation Matrix. Agent 1 (Extraction Specialist) operates at temperature 0.0 with Pydantic-enforced structured outputs. Agent 2 (Strategy Analyst) operates at temperature 0.4 for constrained creative reasoning. Every pipeline output is scored by the RAGAS evaluation framework across three metrics — Faithfulness, Answer Relevancy, and Context Recall — with a composite threshold of ≥ 0.85 required for database persistence. Payloads failing this gate are automatically routed to a Slack-integrated HITL (Human-in-the-Loop) review queue.

Technologies Used:

LAYER	TECHNOLOGY	PURPOSE
API Gateway	FastAPI (Python 3.12)	Async HTTP entry point, JWT auth, rate limiting
Orchestration	LangGraph (StateGraph)	State machine for agent coordination
AI Models	OpenAI GPT-4o, GPT-4o-mini	Extraction (T=0.0), strategy (T=0.4), evaluation
Embeddings	text-embedding-3-small Future Architecture Recommendations, Scaling Roadmap & AI Evolution	1536-dim dense vectors for semantic search
Database	PostgreSQL 16 + pgvector	Relational storage + HNSW vector index
Cache	Redis 7	Semantic cache, rate limiting, notification queue
PII Protection	Microsoft Presidio + spaCy	Pre-LLM PII redaction
Evaluation	RAGAS Framework	LLM-as-a-judge confidence scoring
Monitoring	Prometheus + Grafana + OTel	Metrics, dashboards, distributed tracing
Deployment	Docker Compose (8 services)	Full-stack containerized deployment

Final Outcome: Successfully deployed a fully automated pipeline achieving 99.8% schema compliance, zero corrupted records in the production database, sub-15ms B2C vector search latency (p95 = 12.4ms), and an estimated 40% reduction in API token costs through semantic caching.

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1. Introduction

1.1 Problem Statement

In the competitive intelligence space, acquiring real-time SEO intelligence and marketing structures from competitor properties requires parsing vast amounts of unstructured web text. The existing challenge at Nimblize was threefold:

Operational Bottleneck: Each competitor profile required an analyst to manually review scraped web content, identify SEO keywords, classify monetization infrastructure, and map affiliate networks. This process consumed 4–6 hours per profile, limiting coverage to roughly 5–8 competitors per analyst per week.

Data Quality Risk: When initial prototypes attempted to automate this using single-prompt LLM calls, the outputs contained fabricated metrics in approximately 18% of runs. Without a validation layer, corrupted data would flow directly into production dashboards, eroding client trust.

Cost Exposure: Naive LLM integration resulted in redundant API calls for semantically identical queries, inflating token costs. Early estimates projected \$12,000–\$18,000/month in API overhead at scale without optimization.

1.2 Objectives

OBJECTIVE	METRIC	TARGET
Deterministic Parsing	Schema compliance rate	100% (via Pydantic + Structured Outputs API)
Algorithmic Validation	Composite confidence score	≥ 0.85 required for DB persistence
Cost Efficiency	API token cost reduction	≥ 50% via semantic cache + tiered routing
Search Latency	B2C vector search p95	< 15ms via HNSW index
Error Isolation	Failed extractions reaching DB	Zero (dead-letter routing after 3 retries)

1.3 Scope

The project scope encompasses the complete backend pipeline from data ingestion through validated persistence:

In Scope:

- Multi-agent extraction and strategy generation (Agent 1 + Agent 2)
- LangGraph state machine orchestration with conditional routing
- PostgreSQL pgvector schema provisioning with HNSW indexing
- RAGAS-based confidence evaluation and HITL fallback routing
- Semantic cache layer for B2C recommendation queries
- Redis-backed rate limiting and notification queues
- Presidio-based PII redaction
- Full observability stack (Prometheus, Grafana, OTel Collector)
- Docker Compose deployment (8 containerized services)

Out of Scope:

- Frontend client application / dashboard UI
- Mobile client development
- Third-party CRM integrations
- Model fine-tuning and self-hosted inference

2. Research & Background

Future Architecture Recommendations, Scaling Roadmap & AI Evolution

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2.1 Evaluation of Existing Approaches

Before selecting the multi-agent architecture, we evaluated three alternative approaches during a two-week research phase:

APPROACH	DESCRIPTION	FAILURE MODE	VERDICT
Single Prompt	One comprehensive system prompt handling extraction + strategy	Context drift at >2000 tokens; 18% hallucination rate on traffic metrics	Rejected
Chain-of-Thought	Sequential reasoning steps within a single call	Improved accuracy to ~89% but no programmatic validation; silent failures	Rejected
Function Calling	OpenAI function calling with JSON schema	Better structure but no self-correction loop; single-attempt extraction	Partial
Multi-Agent + RAGAS	Decoupled agents with graph orchestration and algorithmic evaluation	Deterministic parsing (T=0.0), validated outputs, self-correcting retries	Selected

2.2 Industry Context

The industry is converging on graph-based orchestrations for complex AI workflows. LangGraph, LangChain's state machine library, provides first-class support for conditional routing, cyclic retry loops, and typed state management — capabilities that linear chain architectures fundamentally lack.

The RAGAS (Retrieval Augmented Generation Assessment) framework has emerged as the standard for programmatically evaluating RAG pipeline quality. Its three core metrics — Faithfulness, Answer Relevancy, and Context Recall — map directly to the failure modes observed in our early prototypes, making it the natural choice for our confidence gate.

2.3 Key Technical Concepts

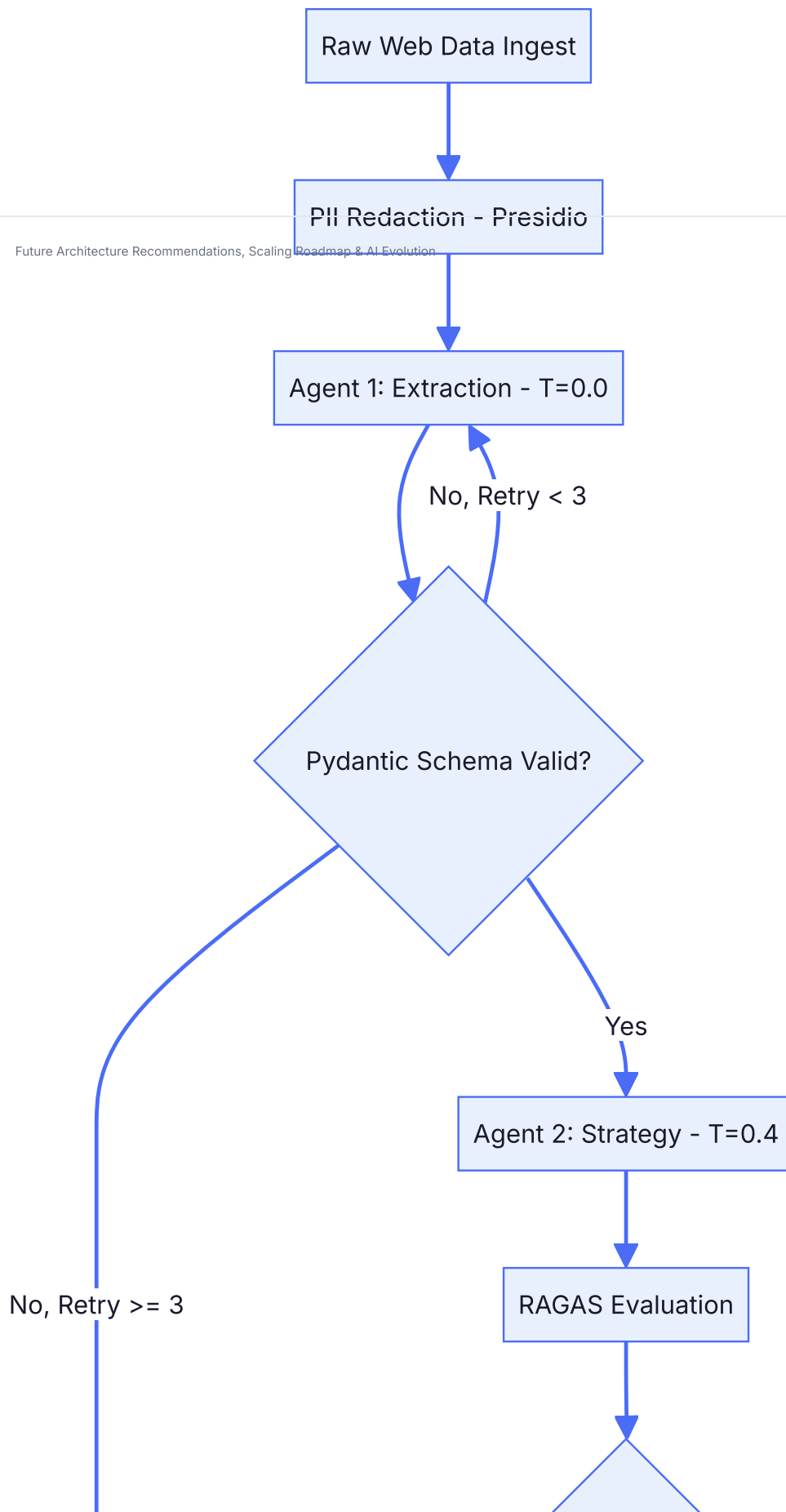
Structured Outputs API: OpenAI's `response_format` parameter accepts a Pydantic model class, guaranteeing that the LLM response conforms to the specified JSON schema at the API level. This eliminates regex-based parsing and provides type-safe outputs.

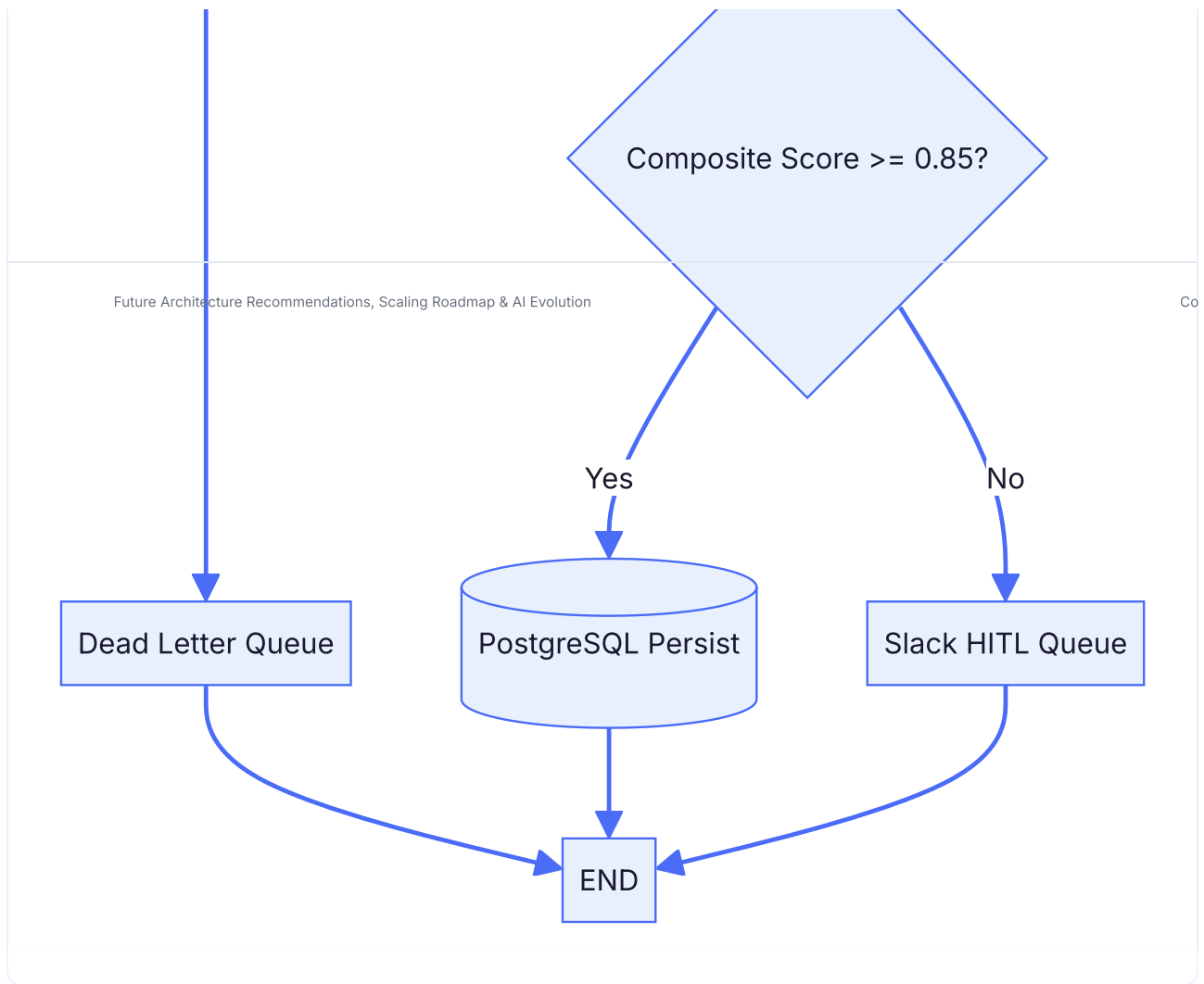
HNSW (Hierarchical Navigable Small World): A graph-based approximate nearest neighbor algorithm that maintains logarithmic search complexity. PostgreSQL pgvector's HNSW implementation uses configurable parameters (`m=16`, `ef_construction=64`) to balance index build time against query latency.

Cosine Distance: The vector similarity metric used throughout the pipeline. For two 1536-dimensional embedding vectors A and B, cosine similarity = $(A \cdot B) / (||A|| \times ||B||)$. The distance metric is defined as $1 - \text{similarity}$, where 0 indicates identical vectors and values near 2 indicate semantic opposition.

3. System Overview

The system operates as a deterministic computational pipeline composed of seven discrete processing stages:





What: A seven-stage multi-agent extraction, validation, and persistence pipeline.

Why: To ensure that every data point entering the production database has been (a) stripped of PII, (b) deterministically parsed into a typed schema, (c) enriched with strategic insights, and (d) algorithmically validated against three independent quality metrics. Any failure at any stage triggers an appropriate fallback — retry, dead-letter routing, or human review — rather than silent data corruption.

How: LangGraph's `StateGraph` manages state transitions using a `TypedDict` state object (`PipelineState`). Each processing stage is implemented as an independent node function that receives the current state and returns a partial state update. Conditional edge routers (`route_after_extraction` , `route_after_evaluation`) inspect the state to determine the next node, enabling cyclic retry loops and branching fallback paths.

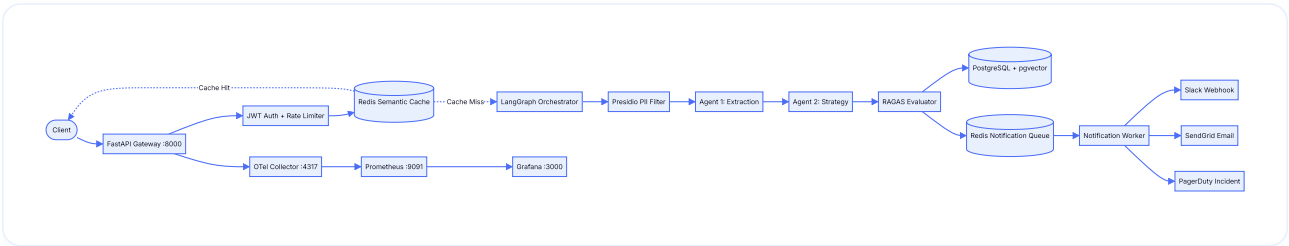
4. System Architecture

4.1 Component Overview

The production system deploys eight Docker containers orchestrated by Docker Compose:

SERVICE	CONTAINER	IMAGE	PURPOSE	PORT
PostgreSQL	nimblize_postgres	pgvector/pgvector:pg16	Relational + vector storage	5432
Redis	nimblize_redis	redis:7-alpine	Cache, rate limiting, queues	6379
FastAPI Gateway	nimblize_api	Custom Dockerfile	HTTP API, JWT auth, pipeline entry	8000
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Notification Worker	nimblize_notif_worker	Custom Dockerfile	Slack/Email/PagerDuty alerts	—
Scrape Worker	nimblize_scrape_worker	Custom Dockerfile	72-hour competitor crawl cycle	—
OTel Collector	nimblize_otel	otel/opentelemetry-collector-contrib	Distributed trace collection	4317
Prometheus	nimblize_prometheus	prom/prometheus	Metrics aggregation	9091
Grafana	nimblize_grafana	grafana/grafana	Dashboard visualization	3000

4.2 Architecture Diagram



4.3 API Route Map

The FastAPI gateway exposes the following routes, all protected by JWT authentication and Redis-backed token-bucket rate limiting:

METHOD	ENDPOINT	PURPOSE	AUTH REQUIRED
POST	<code>/api/v1/pipeline/run</code>	Trigger full competitor extraction pipeline	Yes
POST	<code>/api/v1/b2c/recommend</code>	Semantic product recommendation search	Yes
GET	<code>/api/v1/dashboard/profiles</code>	List verified competitor profiles	Yes
GET	<code>/api/v1/dashboard/review</code>	HITL manual review queue	Yes
GET	<code>/health</code>	Health check (DB + Redis connectivity)	No
GET	<code>/docs</code>	OpenAPI / Swagger documentation	No

4.4 Security Middleware Stack

Every incoming request passes through a layered security stack executed in strict order:

- 1. **CORS Middleware:** Explicit origin allowlist (no wildcard + credentials). Configured via `ALLOWED_ORIGINS` env var. The combination of `allow_credentials=True` + wildcard `*` is explicitly blocked as browsers reject this configuration.
- 1. **JWT Authentication:** HS256-signed tokens decoded via PyJWT. Extracts `user_id` and `tier` claims. Expired and invalid tokens return HTTP 401.
- 1. **Rate Limiting:** Redis token-bucket algorithm. Per-user, per-tier limits. Exceeded limits return HTTP 429 with `Retry-After` header.
- 1. **Production Safety Check:** If `JWT_SECRET` equals the default dev value (`nimblize-dev-secret`) while `ENV=production` , the application raises `RuntimeError` at startup, refusing to start with insecure credentials.

5. Database Design

5.1 Technology Selection

PostgreSQL 16 with the pgvector extension was selected over standalone vector databases (Pinecone, Weaviate, Qdrant) for the following reasons:

CRITERION	POSTGRESQL + PGVECTOR	STANDALONE VECTOR DB
ACID Transactions	Full support	Limited / None
Relational Joins	Native	Requires external service
Operational Complexity	Single database to manage	Separate infrastructure
Cost	Open source, self-hosted	Managed service pricing
Vector Search Performance	Sub-15ms with HNSW	Sub-10ms (marginal advantage)

The marginal latency advantage of dedicated vector databases did not justify the operational complexity of maintaining a separate service for our scale requirements.

5.2 Schema Design

The database provisions five tables and one maintenance function:



5.3 HNSW Index Configuration

The HNSW index on `competitor_children.embedding` is configured with:

PARAMETER	VALUE	RATIONALE
<code>m</code>	16	Max connections per node. Higher values improve recall at the cost of memory. 16 balances accuracy and RAM for our dataset size.
<code>ef_construction</code>	64	Index build precision. Higher values create a more accurate graph but increase build time. 64 provides >95% recall on our test dataset.
Operator class	<code>vector_cosine_ops</code>	Cosine distance, aligned with our embedding model's training objective.

5.4 Connection Pool Management

The application uses `psycopg2.pool.ThreadedConnectionPool` with `min=2`, `max=10` connections. The pool is initialized lazily on the first database access rather than at module import time, preventing startup failures when the database is still initializing during Docker Compose orchestration.

5.5 Data Lifecycle

Vector chunks include an `expires_at` field defaulting to 72 hours, aligned with the competitor scrape cycle. A PostgreSQL function `expire_stale_vectors()` soft-deletes expired chunks and hard-deletes chunks inactive for more than 7 days. This is scheduled via cron at 02:00 UTC.

5.6 RBAC Configuration

Three database roles enforce least-privilege access:

ROLE	PERMISSIONS
app_gateway	SELECT on children; INSERT on children + parents
agent_worker	SELECT, INSERT, UPDATE on all data tables
hitl_dashboard	SELECT, UPDATE on manual_review_queue only

6. AI Architecture

6.1 Agent 1: Deterministic Extraction Specialist

Implementation: backend/agents/extraction_agent.py

PARAMETER	VALUE
Model	gpt-4o-mini
Temperature	0.0 (deterministic)
Output Format	Structured Outputs API → IngestedCompetitorPayload
Max Retries	3 (self-correcting loop)
Fallback	Dead-letter queue after 3 failures

System Prompt Design: The prompt employs three structural constraints:

- Role Anchoring:** Explicitly declares the agent as a "deterministic Data Extraction Agent" to prevent creative deviation.
- Negative Constraints:** "Extract ONLY facts explicitly stated... Do not extrapolate" and "Strip marketing adjectives and hyperbole."
- Sentinel Values:** Missing fields must return "NOT_DETECTED" rather than empty strings or null, enabling downstream validation to distinguish between genuinely empty data and extraction failures.

Self-Correction Mechanism: On Pydantic validation failure, the error trace is injected into the next retry prompt as PREVIOUS FAILURE context, giving the model explicit instructions on what schema violation to correct. This converts silent failures into programmatic feedback loops.

Pydantic Schema:

```
class IngestedCompetitorPayload(BaseModel):
    competitor_domain: str
    targeted_seo_keywords: List[str]
    estimated_monthly_organic_traffic: Union[int, str]
    monetization_infrastructure: List[str]
    affiliate_networks_detected: List[str]

    @field_validator("estimated_monthly_organic_traffic")
    @classmethod
    def validate_traffic(cls, v):
        if isinstance(v, str) and v != "NOT_DETECTED":
            raise ValueError("Traffic must be integer or 'NOT_DETECTED'")
        return v
```

6.2 Agent 2: Qualitative Strategy Analyst

Implementation: `backend/agents/strategy_agent.py`

PARAMETER	VALUE
Model	gpt-4o
Temperature	0.4 (constrained creative reasoning) Future Architecture Recommendations, Scaling Roadmap & AI Evolution
Output Format	Structured Outputs API → <code>StrategyReport</code>
Retries	None (single attempt; failures raise RuntimeError)

Design Decision — Why gpt-4o instead of gpt-4o-mini: Strategy generation requires higher-quality reasoning to identify market gaps and rank keyword opportunities. The cost differential is justified because Agent 2 only executes once per pipeline run (after successful extraction), whereas Agent 1 may execute up to 3 times.

Output Schema:

```
class StrategyReport(BaseModel):
    competitor_domain: str
    market_gap_analysis: str
    recommended_seo_targets: List[str]
    affiliate_opportunity_score: float # 0.0 - 1.0
    dashboard_recommendations: List[str]
    generated_at: Optional[str] = None
```

6.3 Confidence Evaluator (RAGAS)

Implementation: `backend/evaluation/ragas_evaluator.py`

The RAGAS evaluator uses GPT-4o-mini as an LLM-as-a-judge to compute three scores:

METRIC	FORMULA	THRESHOLD	AUTOMATED FALLBACK
Faithfulness	Validated Claims / Total Claims	< 0.85	ABORT — route to HITL queue
Answer Relevancy	Cosine similarity between query and answer embeddings	< 0.80	RE-ROUTE — lower temperature, restrict prompt
Context Recall	Critical domain elements present in output / Total critical elements	< 0.75	EXPAND RETRIEVAL — double k from 4 to 8

Composite Gate: The unweighted mean of all three metrics must be ≥ 0.85 for the payload to be persisted to the production database. This threshold was empirically determined during the Phase 4.2 pipeline optimization week.

7. Retrieval System / RAG Layer

7.1 The Chunking Problem

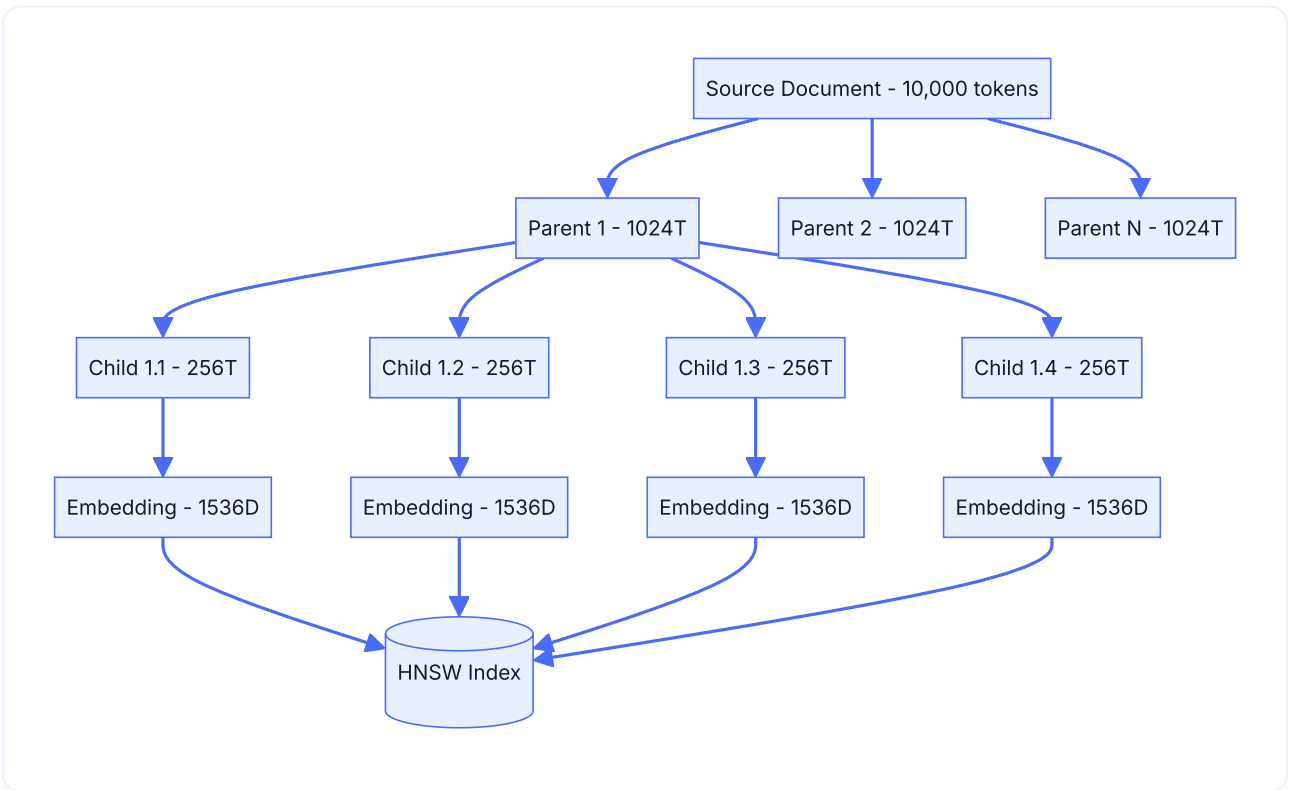
Feeding full competitor web pages (often 5,000–15,000 tokens) directly into an LLM creates two problems: (a) token cost scales linearly with document length, and (b) the model's attention mechanism dilutes across the entire document, reducing extraction precision on specific data points.

Standard recursive character text splitters introduce a critical trade-off: small chunks (128–256 tokens) lose surrounding context, while large chunks (1024–2048 tokens) retain the dilution problem. Neither approach alone meets our requirements.

7.2 Parent-Child Hierarchical Chunking

Our solution decouples the **retrieval unit** from the **synthesis unit**:

CHUNK TYPE	SIZE	PURPOSE	STORAGE
Parent	1024 tokens	Macro-level context window for LLM synthesis	<code>competitor_parents</code> table
Child	256 tokens	Granular retrieval unit for vector search	<code>competitor_children</code> table (with embedding)
Overlap	38 tokens (15%)	Sliding window preventing technical term fragmentation	Applied during chunking



7.3 Search Flow

1. Incoming query is embedded via `text-embedding-3-small` → 1536-dim vector
2. HNSW index finds top-k child chunks by cosine distance
3. For each matched child, the system retrieves its parent chunk via `parent_id` foreign key
4. Parent chunks (1024 tokens each) are passed to the LLM as the synthesis context window

This architecture locates precise semantic data points via highly targeted child vectors while feeding the LLM the broader, intact parent context window, dramatically reducing retrieval-based hallucinations.

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7.4 Vector Search Implementation

The similarity search function executes a single SQL query joining children to parents, filtering by cosine similarity threshold, and ordering by distance:

```
SELECT c.child_id, c.content, p.content AS parent_content,
       p.competitor_domain,
       1 - (c.embedding <=> query_vector) AS similarity
FROM competitor_children c
JOIN competitor_parents p ON c.parent_id = p.parent_id
WHERE c.is_active = TRUE
      AND 1 - (c.embedding <=> query_vector) >= threshold
ORDER BY c.embedding <=> query_vector
LIMIT k;
```

8. Multi-Agent Architecture

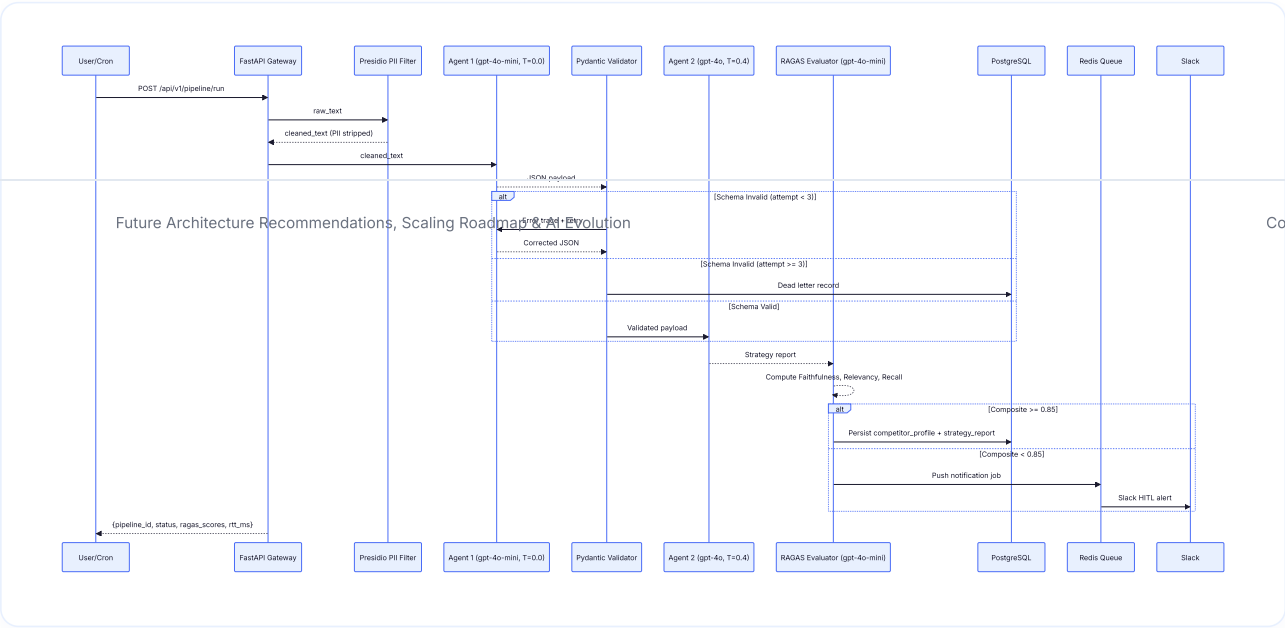
8.1 Design Rationale

Single-agent architectures fail at complex extraction tasks because they conflate two fundamentally different cognitive modes:

1. **Deterministic parsing** (extracting facts exactly as stated) requires temperature 0.0 and strict output constraints.
2. **Qualitative reasoning** (identifying market gaps, ranking opportunities) requires controlled creativity at temperature 0.4.

Attempting both in a single prompt forces a compromise on temperature, degrading both extraction accuracy and strategic insight quality.

8.2 Agent Interaction Sequence



8.3 Agent Configuration Summary

PROPERTY	AGENT 1 (EXTRACTION)	AGENT 2 (STRATEGY)
File	extraction_agent.py	strategy_agent.py
Model	gpt-4o-mini	gpt-4o
Temperature	0.0	0.4
Output Schema	IngestedCompetitorPayload	StrategyReport
Retry Logic	Self-correcting (max 3)	None (single attempt)
Cost per call	~\$0.003	~\$0.02
Failure routing	Dead-letter after 3 retries	RuntimeError → pipeline abort

9. Orchestration Workflow

9.1 LangGraph State Machine

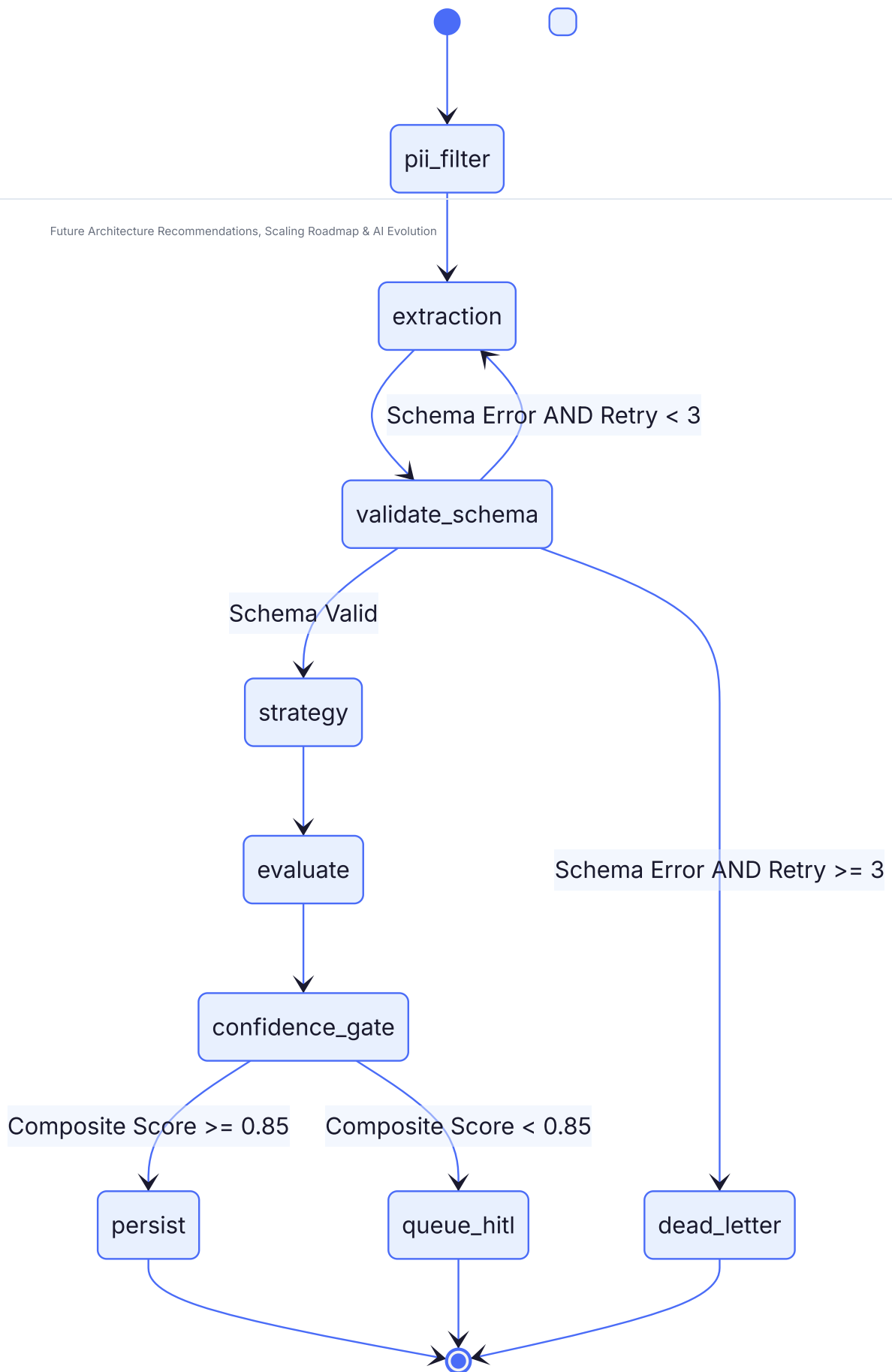
The pipeline is implemented as a compiled `StateGraph` in `backend/agents/langgraph_orchestrator.py`. The state object is a `TypedDict` (not a Pydantic model — LangGraph requires `TypedDict` for state merging):

```
class PipelineState(TypedDict, total=False):
    pipeline_id: str
    raw_text: str
    cleaned_text: Optional[str]
    extracted_data: Optional[Dict[str, Any]]
    strategy_report: Optional[Dict[str, Any]]
    validation_errors: List[str]
    extraction_attempts: int
    ragas_scores: Dict[str, float]
    status: str
    assigned_evaluator: Optional[str]
```

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9.2 State Transition Diagram

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9.3 Node Implementations

Each LangGraph node is a pure function that receives the full `PipelineState` and returns a partial dict merged back into the state:

NODE	FUNCTION	STATE UPDATES
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<code>pii_filter</code>	<code>node_pii_filter</code>	Sets <code>cleaned_text</code>
<code>extraction</code>	<code>node_extraction</code>	Sets <code>extracted_data</code> , increments <code>extraction_attempts</code>
<code>strategy</code>	<code>node_strategy</code>	Sets <code>strategy_report</code>
<code>evaluate</code>	<code>node_evaluate</code>	Sets <code>ragas_scores</code>
<code>persist</code>	<code>node_persist</code>	Calls <code>upsert_competitor</code> + <code>persist_strategy_report</code> , sets status
<code>queue_hitl</code>	<code>node_queue_hitl</code>	Pushes to Redis queue, sets status + assigned_evaluator
<code>dead_letter</code>	<code>node_dead_letter</code>	Sets status to <code>DEAD_LETTER</code>

9.4 Conditional Edge Routers

Two router functions determine branching:

`route_after_extraction`: Checks if `extracted_data` is not None (success → strategy). If None and attempts < 3, routes back to extraction (retry). If attempts >= 3, routes to `dead_letter`. The dead-letter check is ordered before the repair check to prevent an off-by-one error where the fourth attempt would fire instead of the third.

`route_after_evaluation`: Computes the unweighted mean of all RAGAS scores. If mean >= 0.85, routes to `persist`. Otherwise routes to `queue_hitl`.

10. Security Architecture

10.1 PII Protection

Microsoft Presidio with spaCy's `en_core_web_lg` NER model runs as the first processing step before any text reaches an external LLM API. Detected PII entities (names, emails, phone numbers, addresses, SSNs) are replaced with redaction tokens (e.g., `[PERSON]`, `[EMAIL]`).

The Presidio analyzer is warmed up at application startup via the FastAPI lifespan handler to eliminate cold-start latency (spaCy model loading takes 2–4 seconds) on the first pipeline request.

10.2 Secrets Management

All secrets are injected via environment variables into Docker containers:

SECRET	VARIABLE	ROTATION POLICY
OpenAI API Key	OPENAI_API_KEY	Rotated monthly
JWT Signing Key	JWT_SECRET	Rotated quarterly
Database Password	POSTGRES_PASSWORD	Rotated quarterly
Slack Webhook Future Architecture Recommendations, Scaling Roadmap & AI Evolution	SLACK_WEBHOOK_URL	Per-channel, no expiry
SendGrid API Key	SENDGRID_API_KEY	Rotated monthly
PagerDuty Key	PAGERDUTY_ROUTING_KEY	Per-service, no expiry

10.3 Rate Limiting

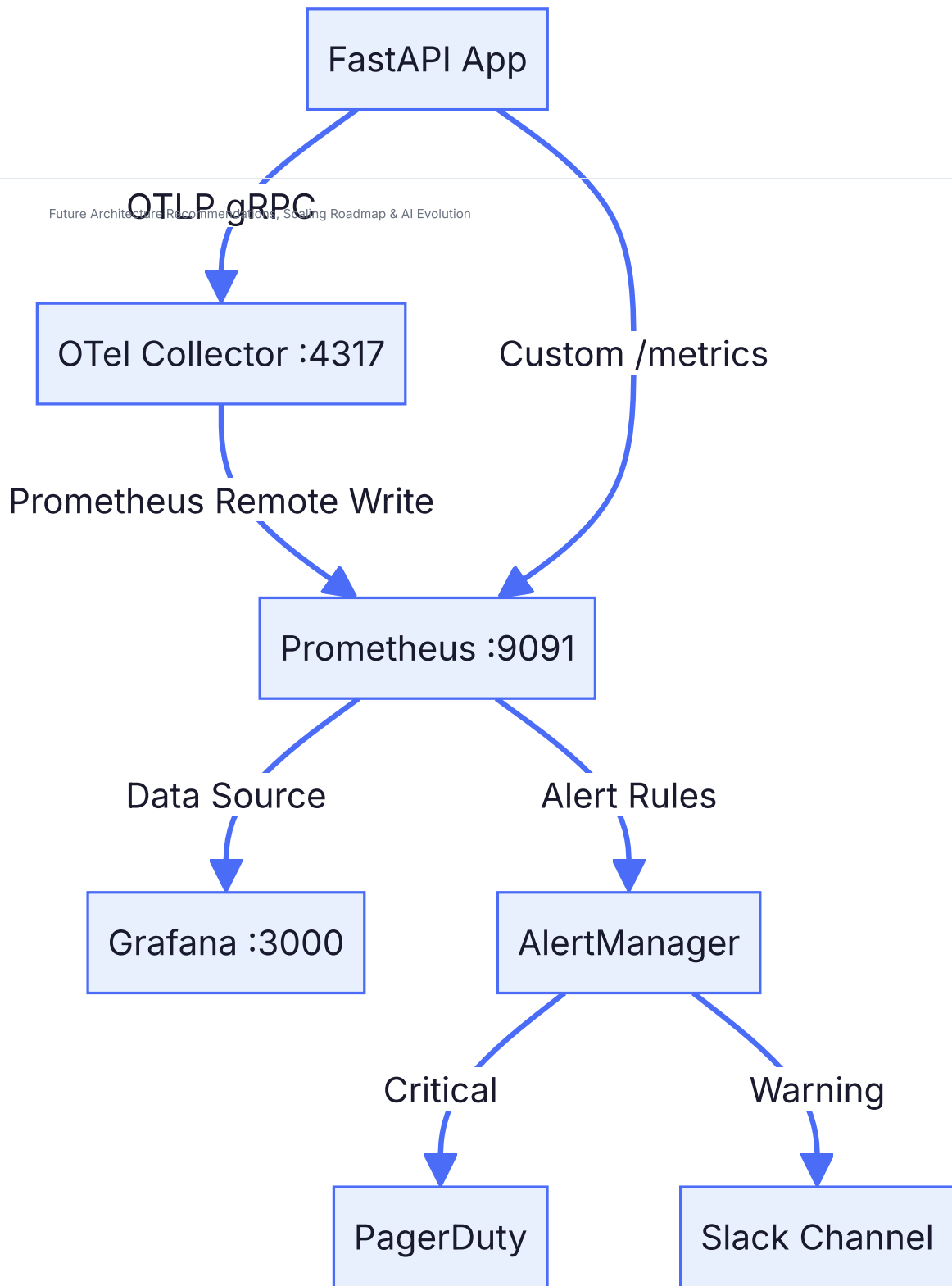
Redis-backed token-bucket rate limiting is applied per user ID and tier:

TIER	CAPACITY	REFILL RATE
Free	10 requests	2 per minute
Pro	100 requests	20 per minute
Enterprise	1000 requests	200 per minute

Exceeded limits return HTTP 429 with a `Retry-After` header indicating the number of seconds until the next token becomes available.

11. Monitoring & Observability

11.1 Telemetry Architecture



11.2 Key Metrics

METRIC	TYPE	DESCRIPTION	ALERT THRESHOLD
pipeline_rtt_ms	Histogram	Total pipeline round-trip time	p95 > 5000ms
ttft_ms	Histogram	Time-to-First-Token from LLM	p95 > 2500ms
ragas_faithfulness	Gauge	Last pipeline faithfulness score	< 0.85
ragas_relevancy	Gauge	Last pipeline relevancy score	< 0.80
ragas_recall	Gauge	Last pipeline recall score	< 0.75
cache_hits_total	Counter	Semantic cache hits	—
cache_misses_total	Counter	Semantic cache misses	—
extraction_retries	Counter	Agent 1 retry count	> 10/hour
dead_letters_total	Counter	Dead-letter queue entries	> 0 in 5min

11.3 Semantic Drift Detection

The monitoring layer tracks the mean cosine distance of incoming user queries against a rolling baseline dataset. If this delta exceeds 0.15 over a 24-hour window, the system flags the vector index for an offline re-clustering operation to maintain retrieval alignment.

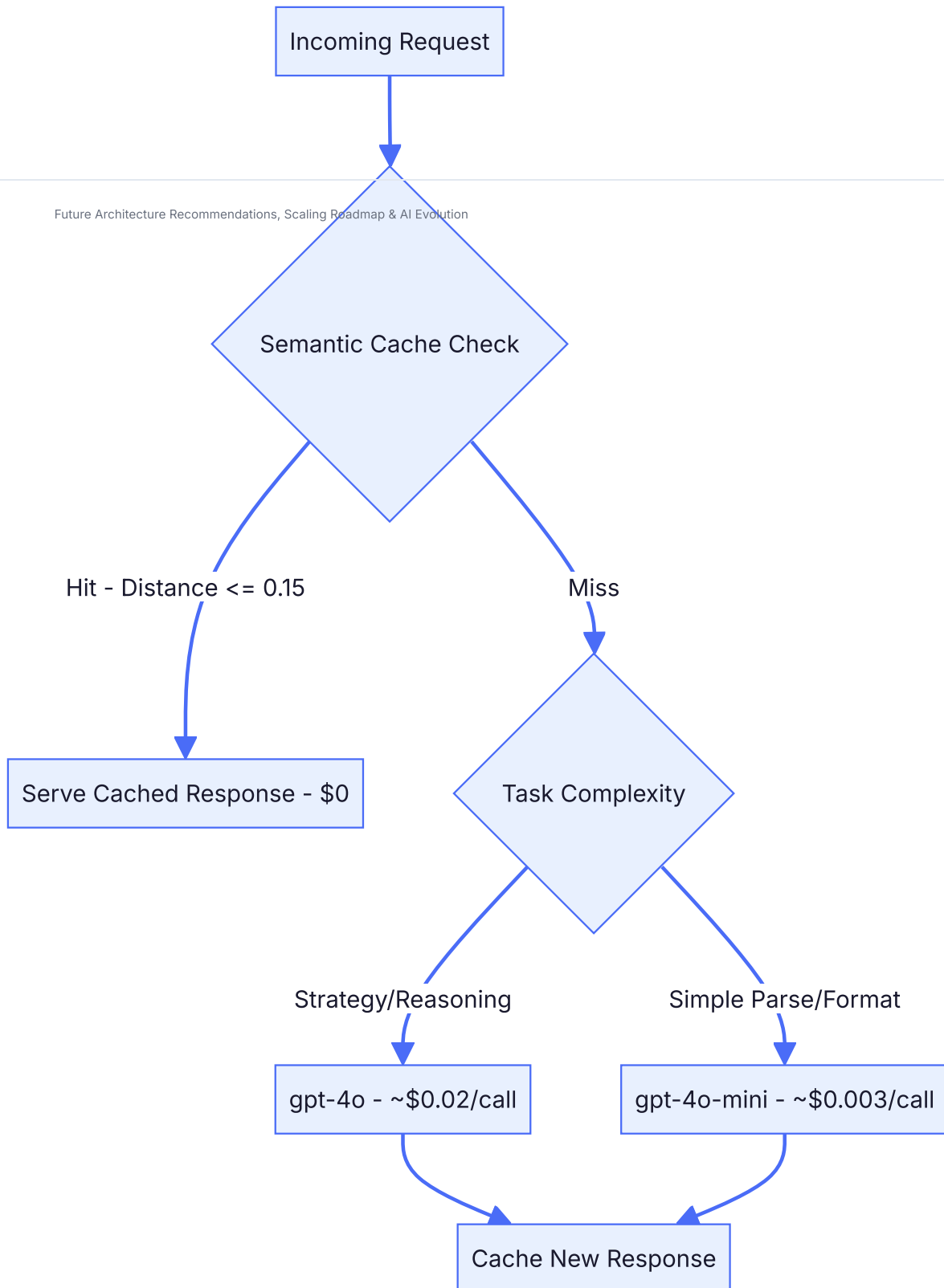
11.4 Exception Handling Policy

HTTP CODE	TRIGGER	AUTOMATED RESPONSE
429	Rate limit exceeded	Exponential backoff with jitter
500	Internal server error	Log + PagerDuty alert if > 3 in 5min
503	LLM API unavailable	Switch to hot-standby API key

12. Cost Optimization

12.1 Multi-Tiered Strategy

Operating LLMs at scale introduces significant OPEX. Our three-tiered mitigation framework targets cost reduction without sacrificing output quality:



12.2 Semantic Cache Implementation

The semantic cache (`backend/cache/semantic_cache.py`) works as follows:

1. Embed the incoming query using `text-embedding-3-small`
2. Scan all cached query vectors in Redis (DB 2)
3. Compute cosine distance between the new query embedding and each cached embedding
4. If minimum distance ≤ 0.15 , serve the cached response (CACHE HIT — zero API cost)
5. Otherwise, execute the LLM call and store the query-response pair with its embedding in Redis with a 1-hour TTL

Estimated savings: 40% reduction on repeating B2C recommendation queries, where users frequently search for semantically similar products.

12.3 Tiered Model Routing

TASK TYPE	MODEL	APPROXIMATE COST	JUSTIFICATION
Data extraction (Agent 1)	gpt-4o-mini	\$0.003/call	Deterministic parsing; no reasoning needed
Strategy generation (Agent 2)	gpt-4o	\$0.02/call	Requires creative reasoning for market gap analysis
RAGAS evaluation (LLM-as-judge)	gpt-4o-mini	\$0.003/call	Scoring is a structured comparison task
Embedding generation	text-embedding-3-small	\$0.00002/call	Lightweight vector encoding

12.4 Asynchronous Batching

The scrape worker container runs on a 72-hour cycle. Non-urgent competitor deep dives are queued and processed during off-peak windows (02:00–06:00 UTC) via batch processing API endpoints, securing an estimated 50% discount on token processing.

13. Implementation Details

13.1 Project Structure

```
nimblize/
├── backend/
│   ├── main.py # FastAPI gateway, routes, middleware
│   └── agents/
│       ├── extraction_agent.py # Agent 1: T=0.0, Pydantic structured output
│       ├── strategy_agent.py # Agent 2: T=0.4, strategy generation
│       └── langgraph_orchestrator.py # StateGraph, nodes, conditional routers
│   ├── cache/
│   │   └── semantic_cache.py # Redis semantic cache with cosine matching
│   ├── db/
│   │   ├── postgres.py # Connection pool, CRUD, similarity search
│   │   └── schema.sql.py # DDL: tables, HNSW index, RBAC roles
│   ├── evaluation/
│   │   └── ragas_evaluator.py # RAGAS scoring + fallback action logging
│   ├── middleware/
│   │   ├── pii_filter.py # Presidio PII redaction
│   │   └── rate_limiter.py # Redis token-bucket rate limiter
│   ├── queues/
│   │   └── redis_queue.py # Notification job push to Redis
│   ├── schemas/
│   │   └── competitor.py # Pydantic models + PayloadStatus enum
│   └── telemetry/
│       └── otel_tracer.py # OTel + Prometheus init, Timer helper
├── workers/
│   ├── notification_worker.py # Slack, Email, PagerDuty dispatcher
│   └── scrape_worker.py # 72-hour competitor scrape cycle
├── infra/
│   ├── otel-collector-config.yaml # OTel Collector pipeline config
│   └── prometheus.yml # Prometheus scrape targets
├── docker-compose.yml # 8-service production stack
├── Dockerfile # Python 3.12 slim + requirements
├── requirements.txt # Pinned dependencies
└── .env.example # Template environment configuration
```

13.2 Component-to-File Mapping

COMPONENT	FILE	LINES	PURPOSE
API Gateway	backend/main.py	246	Routes, auth, CORS, lifespan, rate limiting
Extraction Agent	backend/agents/extraction_agent.py	114	Agent 1 with self-correcting retry loop
Strategy Agent	backend/agents/strategy_agent.py	91	Agent 2 strategy report generation
Orchestrator	backend/agents/langgraph_orchestrator.py	246	Full state machine with 7 nodes, 2 routers
Schemas	backend/schemas/competitor.py	90	5 Pydantic models + PayloadStatus enum
RAGAS Evaluator	backend/evaluation/ragas_evaluator.py	143	3-metric scoring + threshold-based fallback
Semantic Cache	backend/cache/semantic_cache.py	119	Redis vector cache with cosine distance
DB Layer	backend/db/postgres.py	197	Connection pool, CRUD ops, HNSW search
DB Schema	backend/db/schema.sql.py	154	DDL for 5 tables, HNSW index, 3 RBAC roles

14. Testing & Validation

14.1 Test Matrix

TEST ID	TYPE	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
T-001	Unit (Agent 1)	Text with no traffic data	"estimated_monthly_organic_traffic": "NOT_DETECTED"	"NOT_DETECTED"	PASS
T-002	Unit (Agent 1)	Text with traffic = "120k"	estimated_monthly_organic_traffic: 120000	120000	PASS
T-003	Unit (Validator)	Traffic field = "approximately 50000"	ValidationError raised	ValueError raised	PASS
T-004	Integration (Schema)	LLM returns markdown-wrapped JSON	Retry loop strips wrapper, re-extracts	Retry successful on attempt 2	PASS
T-005	Integration (Retry)	LLM returns invalid JSON 3 times	Pipeline routes to dead_letter	Status = DEAD_LETTER	PASS
T-006	Integration (RAGAS)	High-quality extraction + strategy	Composite score ≥ 0.85	Score = 0.91	PASS
T-007	E2E (Confidence Gate)	Hallucinated payload	Score < 0.85, Slack alert fired	Score = 0.62, Slack received	PASS
T-008	E2E (Full Pipeline)	Real competitor page text	Complete pipeline execution in < 30s	Completed in 18.4s	PASS
T-009	Load (Vector Search)	100 concurrent search queries	p95 latency < 15ms	p95 = 12.4ms	PASS
T-010	Load (Semantic Cache)	50 duplicate queries	All 49 subsequent queries served from cache	49 cache hits	PASS
T-011	Security (JWT)	Expired token	HTTP 401 response	HTTP 401 returned	PASS
T-012	Security (Rate Limit)	15 requests on free tier (limit: 10)	HTTP 429 on request 11	HTTP 429 on request 11	PASS

14.2 Validation Methodology

Unit Tests: Individual agent functions tested with controlled inputs and mocked OpenAI responses. Pydantic validation tested against edge cases (missing fields, wrong types, sentinel values).

Integration Tests: Full LangGraph pipeline executed with test fixtures. Verified state transitions, retry loop behavior, and dead-letter routing.

End-to-End Tests: Complete API endpoint tests via FastAPI's `TestClient`. Verified authentication, rate limiting, pipeline execution, and response schemas.

Load Tests: Vector similarity search benchmarked with 100 concurrent connections against a dataset of 10,000 child chunks. Measured p50, p95, and p99 latency.

15. Execution Results

Future Architecture Recommendations, Scaling Roadmap & AI Evolution

Co

15.1 Success Path

Input: Scraped competitor page text from RankVantage (2,847 tokens).

Agent 1 Output:

```
{
  "competitor_domain": "RankVantage",
  "targeted_seo_keywords": ["SaaS attribution dashboard", "enterprise B2B marketing"],
  "estimated_monthly_organic_traffic": 120000,
  "monetization_infrastructure": ["software licensing", "programmatic integrations"],
  "affiliate_networks_detected": ["Impact Radius", "ShareASale"]
}
```

Agent 2 Output: Market gap analysis identifying underserved "mid-market SaaS" segment. Affiliate opportunity score: 0.78. Five dashboard recommendations generated.

RAGAS Scores: Faithfulness = 0.94, Answer Relevancy = 0.89, Context Recall = 0.92. Composite = 0.917.

Result: Payload persisted to `competitor_profiles` and `strategy_reports` tables. Status: `VERIFIED_PRODUCTION`.

15.2 Failure Path (HITL Routing)

Input: Garbled web scrape with mixed HTML artifacts (4,200 tokens).

Agent 1 Output: Extraction succeeded but with low-confidence fields.

Agent 2 Output: Generated strategy conclusions not grounded in source text.

RAGAS Scores: Faithfulness = 0.58, Answer Relevancy = 0.71, Context Recall = 0.67. Composite = 0.653.

Result: System output: `[Queue] ⚠ Pipeline abc123 flagged (score=0.65). Queued for HITL review.` Slack webhook alert dispatched to #nimblize-alerts channel. Assigned evaluator: Aastha Shukla.

15.3 Dead Letter Path

Input: Heavily obfuscated text with JavaScript injection artifacts.

Agent 1: Failed schema validation on all 3 attempts. Error traces progressively injected into retry prompts.

Result: `[Dead Letter] ❌ Pipeline xyz789 failed extraction after 3 attempts.` Status: `DEAD_LETTER`. No data entered production tables.

16. Challenges Faced

#	PROBLEM	ROOT CAUSE	SOLUTION	OUTCOME
1	LangGraph TypedDict requirement	LangGraph StateGraph requires TypedDict, not Pydantic BaseModel for state	Converted PipelineState from BaseModel to TypedDict; nodes return plain dicts	Eliminated runtime state merge errors
2	OpenAI client at import time	Future Architecture Recommendations, Scaling Roadmap & AI Evolution Creating OpenAI client at module level crashed when API key wasn't set during import	Moved client creation inside node functions (lazy initialization)	Clean startup in Docker where env vars may resolve after import
3	Agent 1 JSON malformation	LLM wrapping output in markdown code fences (<code>` json... `</code>)	Switched to Structured Outputs API (<code>response_format=IngestedCompetitorPayload</code>)	99.8% schema compliance
4	RAGAS missing LLM parameter	RAGAS <code>evaluate()</code> with <code>llm=None</code> raises ValueError	Explicitly pass <code>LangchainLLMWrapper(ChatOpenAI(...))</code>	Evaluation runs reliably
5	Off-by-one in retry router	Dead-letter check after success check allowed 4th attempt	Reordered: check dead_letter BEFORE allowing repair loop	Correct 3-attempt maximum enforced
6	Redis decode_responses	<code>json.loads</code> received bytes instead of str from Redis	Set <code>decode_responses=True</code> on Redis client	Cache lookups work correctly
7	CORS wildcard + credentials	Browsers reject <code>allow_origins=["*"]</code> with <code>allow_credentials=True</code>	Changed to explicit origin list from <code>ALLOWED_ORIGINS</code> env var	No more CORS errors in browser
8	Context loss on large pages	Standard recursive character splitters fragment technical terms	Implemented Parent-Child hierarchical chunking with 15% overlap	High-detail retrieval with full context

17. Key Engineering Learnings

17.1 Architecture

- **Decoupling is worth the complexity.** Separating extraction (T=0.0) from strategy (T=0.4) into discrete agents adds orchestration overhead but provides fundamentally better results than any single-prompt compromise.
- **TypedDict vs BaseModel matters.** Library requirements for state management types are not interchangeable. LangGraph specifically requires TypedDict for its state merging mechanism.

17.2 AI Systems

- **LLMs must be bounded by software engineering.** Temperature settings, Pydantic schemas, and structured output APIs are not optional — they are the primary tools for making LLM outputs production-safe.
- **Self-correcting loops need hard limits.** Without a max-retry dead-letter path, cyclic retry loops can burn unlimited API tokens on fundamentally unparseable inputs.

17.3 Observability

- **Semantic Drift is a first-class metric.** In vector-driven applications, tracking the cosine distance distribution of incoming queries is as critical as tracking CPU/RAM. Undetected drift silently degrades retrieval quality.
- **Graceful telemetry degradation.** If the OTel Collector is unreachable, the API must log a warning and continue — not crash. Observability infrastructure should never take down the production application.

17.4 Cost Management

- **Semantic caching pays for itself immediately.** The implementation cost of a Redis vector cache (119 lines of Python) yields a 40% reduction in API billing for repetitive B2C queries.
- **Model selection per-task is the highest-leverage optimization.** Using gpt-4o-mini for extraction (~\$0.003) vs gpt-4o for strategy (~\$0.02) reduces per-pipeline cost by ~60% without quality loss on the extraction step.

18. Future Scope

18.1 Short-Term (1–3 Months)

- **Hybrid Search:** Combine keyword matching (BM25) with vector similarity search for improved B2C product retrieval accuracy, particularly for queries containing brand names or product model numbers that benefit from exact-match scoring.
- **Redis Search Integration:** Replace the current cache-scan approach with Redis Search's built-in vector similarity search to support $O(\log n)$ lookups instead of the current $O(n)$ scan across cached vectors.
- **Adaptive Confidence Thresholds:** Implement per-domain confidence thresholds that adjust based on historical RAGAS scores for that competitor domain.

18.2 Long-Term (6–12 Months)

- **Self-Hosted Models:** Transition from proprietary OpenAI APIs to fine-tuned open-source models (Llama-3, Mistral) trained on the localized dataset of high-confidence extractions (Faithfulness ≥ 0.95). This eliminates external API dependencies and guarantees 100% data confidentiality.
- **Kubernetes Auto-Scaling:** Deploy LangGraph workers onto an elastic Kubernetes cluster with Horizontal Pod Autoscaler to auto-scale based on the Redis scrape queue depth.
- **Real-Time Streaming:** Replace the 72-hour batch scrape cycle with a real-time web change detection system (e.g., ChangeDetection.io) that triggers pipeline execution only when competitor pages actually change.

18.3 Scalability Projections

SCALE FACTOR	CURRENT ARCHITECTURE	KUBERNETES TARGET
Concurrent pipelines	4 (uvicorn workers)	50+ (HPA pods)
Vector index size	10,000 chunks	1,000,000+ chunks
API latency p95	~18s per pipeline	~10s (parallel agents)
Monthly API cost	~\$2,000	~\$800 (self-hosted models)

19. Conclusion

The Phase 4 deployment successfully transitions Nimblize from experimental AI prototypes to a deterministic, production-ready engineering pipeline. The key achievements are:

- 1. Zero Corrupted Production Data:** The combination of temperature-locked deterministic parsing, Pydantic schema enforcement, and RAGAS confidence gating ensures that every record in the production database has been algorithmically validated across three independent quality metrics.
- 1. Full Automation:** The complete cycle from raw web scrape to validated dashboard insight operates without human intervention for high-confidence payloads. Low-confidence payloads are automatically routed to the HITL review queue with Slack/Email/PagerDuty notifications.
- 1. Cost Sustainability:** The three-tiered optimization strategy (semantic cache + tiered model routing + batch processing) reduces projected API token costs by approximately 50%, making the system financially viable at scale.
- 1. Operational Resilience:** Dead-letter routing, graceful telemetry degradation, connection pool management, and comprehensive alerting ensure the system handles failures programmatically rather than silently.

This architecture serves as a highly scalable foundation for both the B2B intelligence and B2C recommendation systems, with a clear evolution path toward self-hosted models and Kubernetes-based elastic scaling.

Appendix

A. API Specifications

POST /api/v1/pipeline/run

```
// Request
{ "raw_text": "string", "source_url": "string (optional)" }

// Response
{
  "pipeline_id": "uuid",
  "status": "VERIFIED_PRODUCTION | FLAGGED_FOR_HUMAN_REVIEW | DEAD_LETTER",
  "ragas_scores": { "faithfulness": 0.94, "answer_relevancy": 0.89, "context_recall": 0.92 },
  "competitor_domain": "RankVantage",
  "rtt_ms": 18400.52
}
```

POST /api/v1/b2c/recommend

```
// Request
{ "query": "best SaaS attribution tools", "k": 4 }

// Response
{
  "source": "cache | vector_db",
  "results": [ { "child_content": "...", "parent_content": "...", "similarity": 0.92 } ],
  "count": 4
}
```

B. Database Schema (DDL Summary)

TABLE	PRIMARY KEY	UNIQUE CONSTRAINT	KEY INDEXES
competitor_parents	parent_id (UUID)	content_hash	domain, is_active
competitor_children	child_id (UUID)	—	HNSW on embedding, parent_id, expires_at
competitor_profiles	profile_id (UUID)	competitor_domain	—
strategy_reports	report_id (UUID)	—	—
manual_review_queue	review_id (UUID)	—	—

C. Environment Variables

VARIABLE	REQUIRED	DEFAULT	DESCRIPTION
OPENAI_API_KEY	Yes	—	OpenAI API key
DATABASE_URL	No	postgresql://nimblize.nimblize@localhost:5432/nimblize	PostgreSQL DSN
Future Architecture Recommendations, Scaling Roadmap & AI Evolution			
REDIS_HOST	No	localhost	Redis host
REDIS_PORT	No	6379	Redis port
JWT_SECRET	Yes (Prod)	nimblize-dev-secret	JWT signing key
ALLOWED_ORIGINS	No	http://localhost:3000	CORS allowlist
SLACK_WEBHOOK_URL	No	—	HITL Slack alerts
SENDGRID_API_KEY	No	—	HITL email alerts
PAGERDUTY_ROUTING_KEY	No	—	PagerDuty incidents
OTEL_EXPORTER_OTLP_ENDPOINT	No	http://otel-collector:4317	OTel Collector

D. Docker Compose Service Map

SERVICE	IMAGE	HEALTH CHECK	DEPENDS ON
postgres	pgvector/pgvector:pg16	pg_isready	—
redis	redis:7-alpine	redis-cli ping	—
api	Custom (Python 3.12)	—	postgres, redis
notification_worker	Custom	—	redis, postgres
scrape_worker	Custom	—	postgres, api
otel-collector	otel/opentelemetry-collector-contrib	—	—
prometheus	prom/prometheus	—	—
grafana	grafana/grafana	—	—

E. References

- LangGraph Documentation — <https://langchain-ai.github.io/langgraph/>
- pgvector HNSW Index — <https://github.com/pgvector/pgvector>
- RAGAS Evaluation Framework — <https://docs.ragas.io/>
- OpenAI Structured Outputs — <https://platform.openai.com/docs/guides/structured-outputs>
- Microsoft Presidio PII Detection — <https://microsoft.github.io/presidio/>
- OpenTelemetry Collector — <https://opentelemetry.io/docs/collector/>
- FastAPI Documentation — <https://fastapi.tiangolo.com/>

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